College of Engineering

EEN210: Digital Circuits

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Project Phase 2

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Abstract:

In this phase, we used Karnaugh maps (K-maps) to simplify the Boolean expressions for a hospital room light control system. The K-maps were developed based on a truth table that we have already accomplished [Phase 1]. By simplifying the Boolean expressions, we managed to find the most possible efficient circuits for the white lights, the yellow light, and the binary LED display, which will make the implementation all the easier.

White lights are on when at least 1 switch is on, behaves like an OR gate.

K-map for white light (W)

S_3S_4	S_1S_2			
	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

Simplified SOP expression for (W):

- $S_1+S_2+S_3+S_4$

Yellow light is ON only when two switches are ON.

K-map for yellow light (Y)

S_3S_4	S_1S_2			
	00	01	11	10
00	0	0	1	0
01	0	1	0	1
11	1	0	0	0
10	0	1	0	1

Simplified SOP expression for (Y):

- $S_1S_2S'_3S'_4 + S'_1S_2S'_3S_4 + S_1S'_2S'_3S_4 + S'_1S'_2S_3S_4 + S'_1S_2S_3S'_4 + S_1S'_2S_3S'_4$

K-map for LED 3

S_3S_4	S_1S_2			
	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	1	0
10	0	0	0	0

- *ON only when all switches are on.*

Simplified SOP expression:

- $S_1S_2S_3S_4$

K-map for LED 2

S_3S_4	S_1S_2			
	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	1	1	0	1
10	0	1	1	1

- *ON when 2/3 switches are on.*

Simplified SOP expression:

- $S_1S_2S'_3 + S_2S'_3S_4 + S_1S'_2S_4 + S'_1S_3S_4 + S'_1S_2S_3 + S_1S_3S'_4$

K-map for LED 1

S_3S_4	S_1S_2			
	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

- *ON only when 1 OR 3 switches are on.*

Simplified SOP expression:

- $S'_1S_2S'_3S'_4 + S_1S'_2S'_3S'_4 + S'_1S'_2S'_3S_4 + S_1S_2S'_3S_4 + S'_1S_2S_3S_4 + S_1S'_2S_3S_4 + S'_1S'_2S_3S'_4 + S_1S_2S_3S'_4$